COMPUTER RESEARCH AND MODELING
2026 VOL. NO. P. 1–19

MATHEMATICAL MODELING AND NUMERICAL SIMULATION

UDC: 519.85:656.13

Origin–destination matrix recovery from partially observed equilibrium link flows

I. V. Podlipnova[1,2,a], E. A. Ilin[1], I. D. Zabara[1], D. V. Yarmoshik[3], E. V. Gasnikova[1,b]

[1]Moscow Institute of Physics and Technology, 9 Institutskiy per., Dolgoprudny, Moscow Region, 141701, Russia
[2]Innopolis University, 1 Universitetskaya st., Innopolis, Russia
[3]Moscow indipendent research institute of artificial intelligence

E-mail: [a] podlipnova.iv@phystech.edu, [b] egasnikova@yandex.ru

An origin–destination (OD) matrix specifies trip volumes between pairs of transportation zones and provides the demand input for equilibrium network assignment. Determining it directly requires detailed information about movements between zones, whereas in practice flow measurements are usually available only on selected links. We study the inverse problem of recovering an OD matrix from such partial observations. The dependence of equilibrium flows on demand is defined implicitly by the convex Beckmann problem with BPR travel-time functions. At the upper level, the objective combines squared flow mismatch on observed links with an entropy regularizer. Total departures and arrivals are treated as known and imposed as hard constraints.

To compute the gradient, we first solve the Beckmann problem and then form a linearized traffic-equilibrium problem. A sparse optimality system is assembled on the shortest-path subgraphs, and its solutions for coordinate OD perturbations form the full Jacobian of equilibrium link flows with respect to demand. The upper-level problem is solved by Adam with a decaying step size. After every iteration, iterative proportional fitting restores nonnegativity, the zero diagonal, and the given marginal totals.

The algorithm is evaluated on the Sioux Falls network under an idealized protocol: the observations are synthesized by the same equilibrium model without noise, and the initial estimate is a perturbed version of the true matrix, so the experiments assess refinement of an informed initial estimate from partial observations. We consider random sets of observed links, several ways of placing a fixed number of sensors, and observation of links carrying the largest flows. Performance is measured by the flow error on the original network and by the mean error after deleting a link while preserving connectivity. The latter metric characterizes transfer under a change in topology. With sparse coverage, reducing mismatch on observed links does not guarantee accurate flows on the entire network. Broader coverage improves recovery, but transfer to a modified topology remains more difficult. Observing heavily loaded links is effective under a limited budget, although selection based only on flow magnitude is not universally preferable. External diagnostic metrics are therefore necessary, and recovery quality depends on both the number and the location of sensors.

Keywords: OD matrix, traffic assignment, bilevel optimization, sensor placement